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ORIGINAL ARTICLE – LABORATORY AND TRANSLATIONAL RESEARCH

Chemoprotective Effects of *Curcuma aromatica* on Esophageal Carcinogenesis

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ABSTRACT Previous studies have demonstrated a decrease in manganese superoxide dismutase (MnSOD) in both Barrett's epithelium of patients and columnar esophageal epithelium of rats after esophagoduodenal anastomosis (EDA). *Curcuma aromatica*, an herbal medicine, has been shown to display anti-carcinogenic properties in a wide variety of cell lines and animals. This study was designed to investigate the ability of *Curcuma aromatica* oil for the prevention of BE and EAC, possibly through its ability to preserve MnSOD function. EDA was performed on rats and *Curcuma aromatica* oil was administered by i.p. injection. Histological changes and oxidative damage were determined after EDA of 1, 3, and 6 months. MnSOD protein level and MnSOD enzymatic activity were evaluated. Lipid peroxidation was determined by TBARs assay and 8-hydroxy-deoxyguanosine for DNA oxidative damage was measured by immunohistochemical staining. In addition, the indexes of both apoptosis and proliferation were determined by PCNA staining and TUNEL assay, respectively. Severe esophagitis were seen in EDA rats, and morphological transformation within the esophageal epithelium was observed with intestinal metaplasia and EAC identified after 3 months. The EDA rats treated with *Curcuma aromatica* oil showed that both MnSOD enzymatic activity and protein level were similar to sham controls. Decreased incidences of intestinal metaplasia and EAC also were observed in the EDA rats with *Curcuma aromatica* oil treatment. *Curcuma aromatica* oil prevented loss of MnSOD in EDA rat esophageal epithelium, and this preservation of MnSOD is associated with the potential protective mechanism against transformation of esophageal epithelial to BE to EAC.

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Previous studies using an external esophageal perfusion rat model[1–3] and a esophagoduodenal anastomosis (EDA) rat model[4] have demonstrated that oxidative damage is closely related to reflux esophagitis and the transition from esophagitis to Barrett's esophagus (BE) to esophageal adenocarcinoma (EAC). In the esophagus we have demonstrated significant increased levels of lipid peroxidation and 8-hydroxy-deoxyguanosine (8-OH-dG), and antioxidants, such as MnSOD and glutathione (GSH), are depressed in the rat esophagi with external esophageal perfusion and in the EDA model at a very early stage. Studies have reported that reactive oxygen species (ROS) can cause DNA injury, such as strand breakage, alterations in guanine and thymine bases, and DNA cross-linkage.[5] Oxidative DNA damage may contribute to the accumulation of genetic damage, and the accumulation of genetic and epigenetic aberrations produces one or more clones with metaplastic and/or malignant potential.[6]

Curcuma aromatica, in Latin botanical known as *Curcuma phaeocaulis* Valeton; *Curcuma kwangsiensis* by Lee et al., or *Curcuma wenyujin* by Chen et al.[7] has been described as a potent antioxidant.[8] Supplementation with *Curcuma longa* reduces oxidative stress and lower plasma lipid peroxide in rabbits fed a high cholesterol diet.[9] A significant reversal in lipid peroxidation, brain lipids, and enhancement of glutathione by *Curcuma aromatica* also was observed in model of brain injury with ethanol intoxicated rats.[10] In addition, the curcuma extract may act as a co-antioxidant that potentiates the effects of antioxidants, such as vitamin E.[11] As a potent antioxidant, curcumin, a cucuminoid ingredient extracted from *Curcuma aromatica* has been widely studied and displayed anti-carcinogenic properties in a wide variety of cell lines and animals.[12–14] *Curcuma aromatica* oil is a volatile oil extracted from *Curcuma aromatica* containing a different spectrum of components related to sesquiterpenoids rather than cucuminoids, and functions as an anti-inflammation, anti-virus, and anti-oxidation agent.[15] The anti- carcinogenic properties of *Curcuma aromatica* oil also have been identified[16]; however, the precise molecular mechanisms underlying its actions against cancer are largely unknown, and there have been no reports of its chemoprotective effects in esophageal adenocarcinoma.

We postulate that oxidative stress is a driving force in the progression of BE and EAC. Oxidative damage is caused by ROS, which induces DNA injury, and accumulation of genetic alternation leads to uncontrolled cell replication, which increases the malignant potential. The inadequate expression of MnSOD suggests that antioxidative defense system is compromised, which exacerbates GERD condition and leads to sever damage to the esophageal epithelium. Therefore, we hypothesized that oxidative stress induces DNA damage, which drives the metaplastic and carcinogenetic transformation in the esophageal epithelium. *Curcuma aromatica* oil protects the esophagus from esophageal reflux injury through its ability to maintain mitochondrial function, leading to MnSOD preservation and prevention against BE and EDA.

The esophagoduodenal anastomosis (EDA) in rats is the most commonly used surgical model to produce duodenogastroesophageal reflux. This model produces reflux esophagitis, metaplastic columnar cell-lined esophagus, and EAC.[17] In this study, we proposed to investigate possible mechanisms by which *Curcuma aromatica* oil protects against oxidative injury in esophageal epithelium of EDA rat during the transformation of BE and EAC.

MATERIALS AND METHODS

Animals and Treatment

Eight-week old Sprague-Dawley rats (Harlan, Indianapolis, IN) were housed three per cage, given commercial rat chow and tap water, and maintained on a 12-hour light/dark cycle. They were allowed to acclimate for 2 weeks before surgery. Solid food was withdrawn 1 day before and 1 day after surgery. EDA was performed on rats according to the operating procedure described previously.[18] This study was approved by the Institutional Animal Care and Use Committee at the University of Louisville. Postoperatively, the rats were given water after 2 hours and rat chow the following day. After surgery rats were treated with *Curcuma aromatica* oil, which is extracted from *Curcuma wenyujin* by supercritical fluid extraction and trapping with silica gel column (School of Pharmacy, Wenzhou Medical College, China). Rats were dosed with *Curcuma aromatica* oil at 100 mg/kg, i.p. every 3 days, and the same volume saline as controls. The animals were weighed weekly. Rats were killed after EDA at 1, 3, and 6 months.

Histopathology

The entire esophagus was removed and opened longitudinally to examine for evidence of gross abnormalities. The samples of esophageal tissues (0.5 cm in length) were taken from the distal part of the esophagus and fixed in 10% buffered formalin for 24 hours and transferred to 80% ethanol. The formalin-fixed esophagus was embedded in paraffin. Serial sections of 5-μm were mounted onto glass slides for histopathological and immunohistochemical analysis. Hematoxylin and eosin (H&E)-stained slides were obtained for each rat. Evidence for reflux esophagitis was identified in the esophageal epithelium, such as the infiltration of inflammatory cells, basal cell hyperproliferation, papillae hypertrophy, dilation of venules, in-growth of the capillaries, epithelial sloughing, and ulceration.[19]

TUNEL Assay

ApopTag® in situ apoptosis detection kit (Intergen Company, Purchase, NY) was used to detect the apoptotic cells according to a procedure reported previously.[20,21] The TUNEL-positive epithelial cells were counted against negative cells under a light microscope at a magnification of ×40; six visual epithelium fields were chosen on each slide, and all sections from each animal were examined. An apoptotic index (number of epithelial nuclei labeled by the TUNEL method/ number of total epithelial nuclei) was calculated.

Immunohistochemical Assay

Immunohistochemical assays were performed to detect proliferating cell nuclear antigen (PCNA) and 8-hydroxydeoxyguanosine (8-OH-dG). Immunohistochemical staining was performed on the paraffin-embedded material using the DAKO EnVision™ +System Kit (DAKO Corporation, Carpinteria, CA). PCNA-positive epithelial cells are counted against negative cells under a light microscope at a magnification of ×40. Same as apoptotic index, six visual fields are chosen on each slide, proliferation index is calculated as a ratio of the number of PCNA-positive epithelial nuclei and the number of total epithelial nuclei. The digital images of 8-OH-dG staining is acquired with the microscope at ×40 magnification using the Spot camera via the MetaMorph® Imaging System (Universal Imaging Corporation., Downingtown, PA) and stored as JPG data files (resolutions were fixed as 200 pixels/inch).

The procedure for the computer image analysis was performed, and the acquired color images from the immunohistochemical staining were defined a standard threshold according to the software specification. The computer program quantified the threshold area represented by color images. 8-OH-dG levels were defined by the percentages of threshold area in acquired color images.

Thiobarbituric Acid Reactive Substances (TBARS) Assay

Lipid peroxidation is quantified by an OXItek TBARS Assay Kit (ZeptoMetrix Corporation, Buffalo, NY) measuring the malondialdehyde (MDA) concentrations as described in the provided instruction.

Western Blot Analysis of MnSOD Expression

Western bolt is performed to determine the MnSOD protein expression in the esophageal mucosal layer and muscle layer. In brief, total protein is isolated from fresh tissue samples by homogenization in ice cold buffer containing 20 mM HEPES (pH 7.5), 1.5 mM $MgCl_2$ mM dithiothreitol, 0.4 M NaCl, 20% glycerol, 0.5 mM phenylmethylsulfonyl fluoride, and 0.5 mM leupeptin at 4°C. Insoluble cellular material is removed by microcentrifugation at 16,000 g for 5 min, and total protein is determined spectrophotomertrically. The protein samples are separated via SDS/PAGE and subsequently transferred to the nitrocellulose membrane for Western blot described previously.[22]

SOD Enzymatic Activity Assay

SOD activity is determined by a SOD Assay Kit-WST (Dojindo Molecular Technologies, Inc, Gaithersburg, MD) according to the provided instruction. In brief, this kit allows highly sensitive SOD assay by utilizing a highly water-soluble tetrazolium salt, WST-1 (2-(4-iodophenyl)-3-(4-nitrophenyl)-5-(2,4-disulfo-phenyl)-2H-tetrazolium, monosodium salt), which produces a water-soluble formazan dye upon reduction with a superoxide anion. Samples are tested and a standard curve ranging from 0.156 to 20 U/ml is prepared. The colorimetric assay is performed measuring formazan produced by the reaction between WST-1 and superoxide anion (O^{2-}); the rate of the reduction with O^{2-} is linearly related to the xanthine oxidase activity and is inhibited by SOD. The absorbance was obtained with a microplate reader reading at 450 nm. MnSOD activity is determined by adding 1 mM KCN to samples to block Cu/Zn-SOD activity completely, and then subtract the Cu/ZnSOD activity from total SOD activity.

Glutathione Assay

Glutathione is determined by a Cayman's GSH assay kit (Cayman Chemical, Ann Arbor, MI) utilizing an enzymatic recycling method. Measurement of the absorbance of TNB at 412 nm provides an accurate estimation of GSH in the sample. Because of the use of glutathione reductase in this assay, both GSH and GSSG are measured and the assay reflects total glutathione.

Catalase Assay

Catalase was determined by a Cayman's GSH assay kit (Cayman Chemical) utilizing the peroxidatic function of Catalase for determination of enzyme activity. Measurement of the absorbance at 540 nm provides an accurate estimation of Catalase enzymatic activity in the sample.

Statistical Analysis

Student's t tests assuming unequal variance was performed. The results are expressed as mean values ± standard deviation. Comparisons were made among the bile perfusion groups and saline control groups by analysis of variance. $p < 0.05$ was considered statistically significant.

RESULTS

Esophageal Pathogenesis

All animals that underwent the EDA procedure completed the study. After EDA, the gross appearance of distal esophagus was dilated, mucosal layer was thickened and friable, and the mucosal surface was irregular compared with the nonoperated controls. The abnormal changes were more extensive with three-fourths of distal esophagus involved at EDA after 6 months duration (Fig. 1). Histologically, all rats after EDA were diagnosed with esophagitis, and the degree of esophagitis was increased along with the experimental duration. *Curcuma aromatica* oil treatment significantly reduced the effects of EDA induced reflux esophagitis in the later stages, with squamous hyperplasia and extension of the lamina propria papillae within the esophageal mucosa being ameliorated in the *Curcuma aromatica* oil treated EDA rats from 3 and 6 months (Table 1). Intestinal metaplasia was demonstrated in 50% of the EDA rats, with the finding of columnar cells in the mucosa and submucosa (Table 1). EAC also was demonstrated in the 6 month EDA duration animals; however, in the *Curcuma aromatica* oil treatment EDA rats there was a significant decrease in intestinal metaplasia incidence and no incidences of EAC.

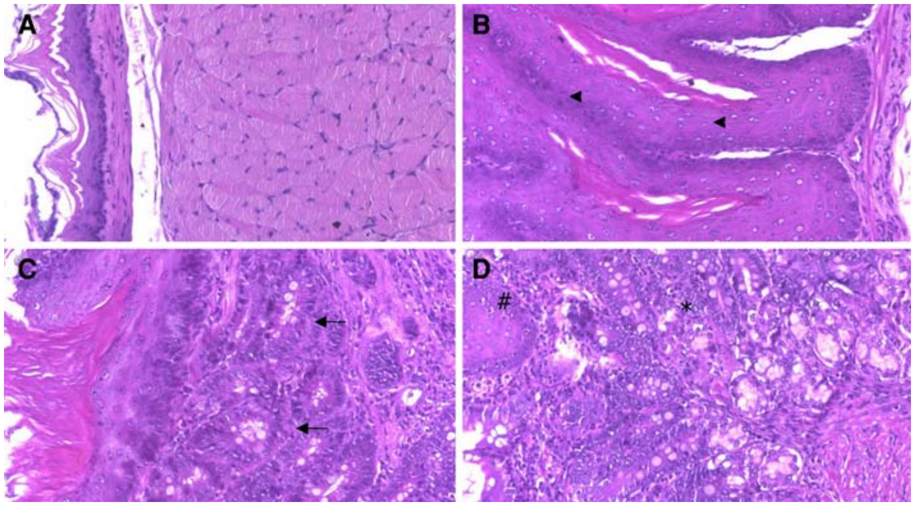

FIG. 1 Representative histology of esophagitis, metaplasia, and EAC. **a** Normal esophageal epithlium. **b** Papillae elongation and mucosa ulceration were seen in the esophageal epithelium after EDA. Papillomatosis with multilayer squamous cells predominated (arrow head). **c** Specific intestinal metaplasia with identification of goblet cells showed in the distal esophageal epithelium (arrow); some squamous epithelium also can be seen. **d** EAC was detected with the adenocarcinoma invading into the muscular layer together with goblet cells in the submucosa (*). Hematoxylin and eosin staining (×200)

TABLE 1 Effects of *Curcuma aromatica* oil on histological changes of rat esophagi after EDA

Group	Month	*n*	No. of animals (incidence %)		
			Esophagitis	Intestinal metaplasia	EAC
Non-OP	1	3	–	0	0
Non-OP	3	3	–	0	0
Non-OP	6	3	–	0	0
EDA	1	10	1.3 ± 0.15	0	0
EDA	3	10	3.3 ± 0.37	3 (30)	0
EDA	6	10	3.4 ± 0.53	7 (70)	4 (40)
EDA + CAO	1	10	1.2 ± 0.33	0	0
EDA + CAO	3	10	1.1 ± 0. 16$^{\#}$	0	0
EDA + CAO	6	10	1.3 ± 0.12$^{\#}$	1 (10)	0

The Hetzel grading system is as follows: 0, normal appearing mucosa; grade 1, mucosa edema hyperemia and/or friability of mucosa; grade 2, superficial erosions involving 10–50% of the esophageal squamous; grade 4, deep peptic ulceration anywhere in the esophagus or confluent erosion of >50% of the esophageal squamous mucosa. Data represent means ± SD of all animals with esophagitis. *Non-OP* nonoperation, *EDA* esophagoduodenal anastomosis, *CAO Curcuma aromatica* oil. Data are mean ± SD. $^{\#}$ $P < 0.05$ compared with group of EDA month 3 and EDA month 6

Proliferation and Apoptosis of Esophageal Epithelial Cells

Esophageal epithelial cell apoptosis and proliferation were assessed under light microscopy. The apoptotic index by counting positive TUNEL staining epithelial cell in EDA esophageal epithelium significantly increased in continuity compared with the nonoperation controls ($p < 0.05$). In the EDA rats with *Curcuma aromatica* oil treatment, a decreased apoptotic index was found after EDA 1 month compared with the saline-treated EDA rats. However, the apoptotic indexes in the EDA rats with *Curcuma aromatica* oil treatment after EDA month 3 and month 6 were significantly increased compared with that in the EDA rats without *Curcuma aromatica* oil treatment ($p < 0.05$). The apoptotic index is shown in Fig. 2a. Proliferating cell nuclear antigen (PCNA) staining was used as an index of the esophageal mucosa proliferation. A number of PCNA-positive cells were seen within the esophageal epithelium, especially in the basal layer in rats after EAD. The proliferating index was significantly increased in the EDA esophageal mucosa compared with the nonoperation controls ($p < 0.05$). In the EDA rats with *Curcuma aromatica* oil treatment, the proliferation index of rats after month 6 was significantly decreased compared with that in the EDA rats without *Curcuma aromatica* oil treatment ($p < 0.05$; Fig. 2b).

Oxidative Damage in the Rat Esophagus

Two measurements regarding oxidative damage were performed in the rat esophageal epithelium after EDA: 8-OH-dG measurement for oxidative damage to DNA, and TBARS measurement for lipid peroxidation. The levels of 8-OH-dG in the esophageal epithelium were evaluated by immunohistochemical staining along with computer image analysis. Compared with the nonoperated controls, the level of 8-OH-dG was significantly increased in the esophageal epithelium of EDA rats. However, the level of 8-OH-dG was not increased in the esophageal epithelium

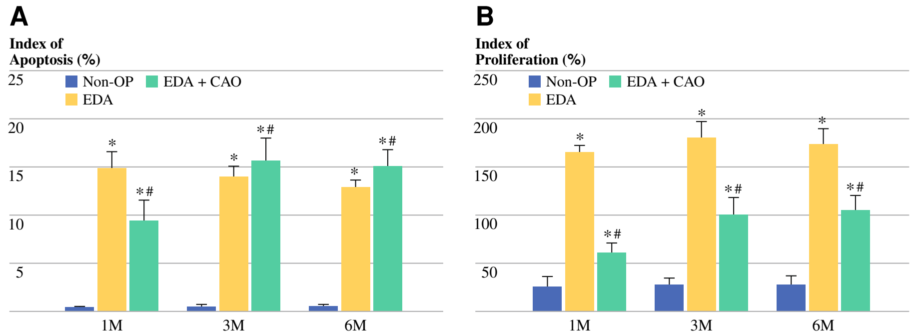

FIG 2 Effect of *Curcuma aromatica* oil on apoptosis and proliferation in rat esophageal epithelium after EDA. **a** Apoptosis by TUNEL assay. **b** PCNA by immunohistochemistry. *Non-OP* nonoperation, *EDA* esophagoduodenal anastomosis, *CAO Curcuma aromatica* oil, *M* month. Data are mean ± SD. * $p < 0.05$ compared with non-OP group; # $p < 0.05$ compared with EDA group

of EDA rats treated with *Curcuma aromatica* oil. The level of lipid peroxidation was considerably increased in rats after EDA compared with nonoperation rats (Fig. 3). Treatment with *Curcuma aromatica* oil resulted in a significant decrease in the levels of lipid peroxidation products in the EDA rats from month 1; however, the levels of lipid peroxidation levels were significantly increased in the *Curcuma aromatica* oil-treated EDA rats from month 3 and month 6 compared with the EDA rats without *Curcuma aromatica* oil treatment from same time points.

Antioxidants in the Rat Esophagus

MnSOD expression and SODs enzymatic activity from the esophageal epithelial cells were dramatically decreased in the rats after EDA 1 month compared with the nonoperated; however, there were significantly increased levels of MnSOD in the EDA rats treated with *Curcuma aromatica* oil. Regarding SODs enzymatic activities, there was a significant loss of total SOD activity in esophageal epithelium of EDA rats compared with that of nonoperation animals. Specific MnSOD enzymatic activity contributed to the loss of total SOD activity, because Cu/ZnSOD enzymatic activity was unaffected. The loss of MnSOD in EDA esophageal epithelium was ameliorated by *Curcuma aromatica* oil treatment ($p < 0.05$; Fig. 4). At EDA 3 month and 6 month, both MnSOD protein level and the enzymatic activity of MnSOD were not decreased, and there was no significant difference among the study groups (data not shown).

The measurements of levels of GSH and catalase enzymatic activities also were performed in the mucosal layer stripped from esophageal tissue. The levels of GSH were significantly lowered in the esophageal epithelium of EDA rats in all stages compared with nonoperation controls; however, treatment with *Curcuma aromatica* oil in EDA rats did not augment the levels of GSH ($p > 0.05$; Fig. 5). The catalase enzymatic activities also were observed in the EDA rats; however, no significant difference was found among all three study groups ($p > 0.05$).

DISCUSSION

To our knowledge, we report the first evidence that *Curcuma aromatica* oil—a volatile oil extracted from *Curcuma aromatica*—has properties that may be useful for the prevention of BE and EAC. We observed that *Curcuma aromatica* oil ameliorated the degree of esophagitis and

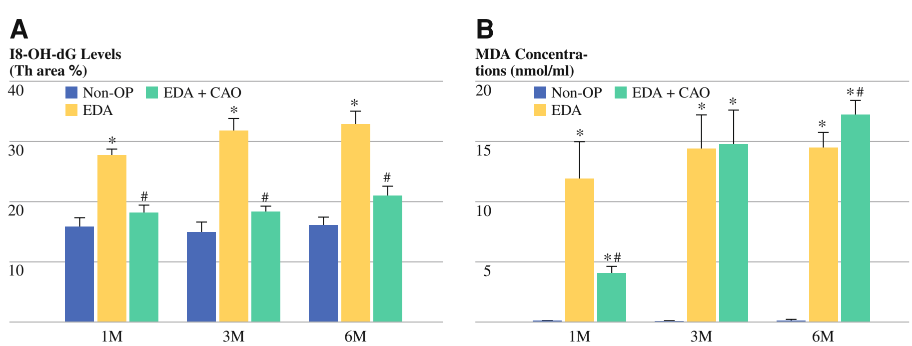

FIG. 3 Effect of *Curcuma aromatica* oil on lipid peroxidatin and 8-OH-dG in rat esophageal epithelium after EDA. **a** 8-OH-dG level by immunohisochemistry. **b** Lipid peroxidatin by (TBARS) assay. *Non-OP* nonoperation, *EDA* esophagoduodenal anastomosis, *CAO Curcuma aromatica* oil, *M* month. Data are mean ± SD. * $p < 0.05$ compared with non-OP group; # $p < 0.05$ compared with EDA group

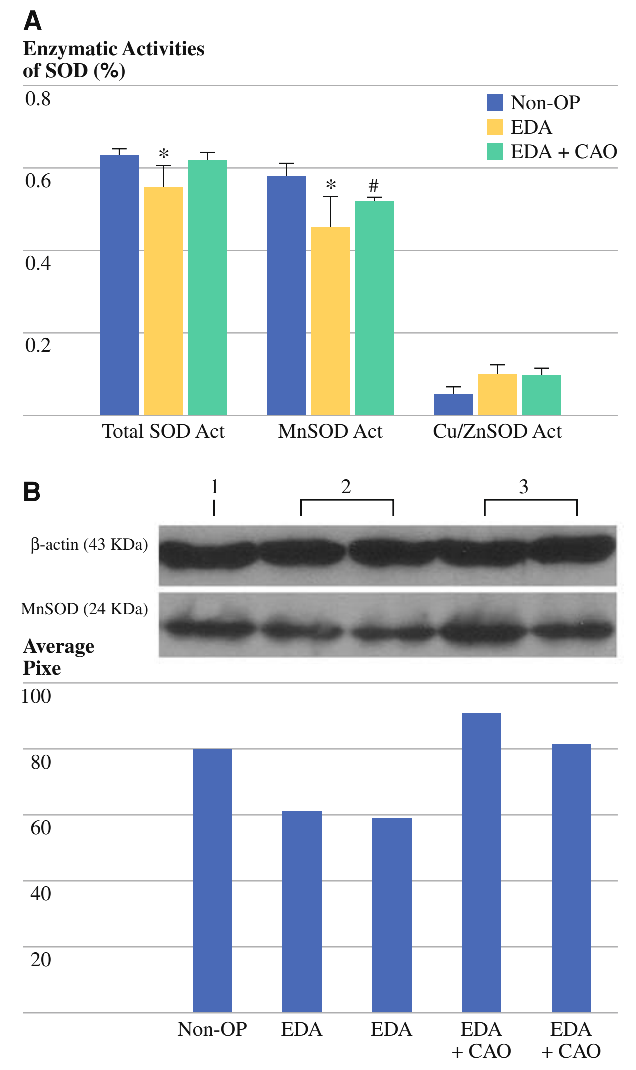

FIG. 4 Effect of *Curcuma aromatica* oil on SODs enzymatic activities and MnSOD expression in rat esophageal epithelium after EDA. **a** SODs enzymatic activities. Data are mean ± SD. * $p < 0.05$ compared with non-OP group; $^{\#}$ $p < 0.05$ compared with EDA group. **b** MnSOD expression. 1 non-OP; 2 EDA; 3 EDA + CAO. *Non-OP* nonoperation, *EDA* esophagoduodenal anastomosis, *CAO Curcuma aromatica* oil

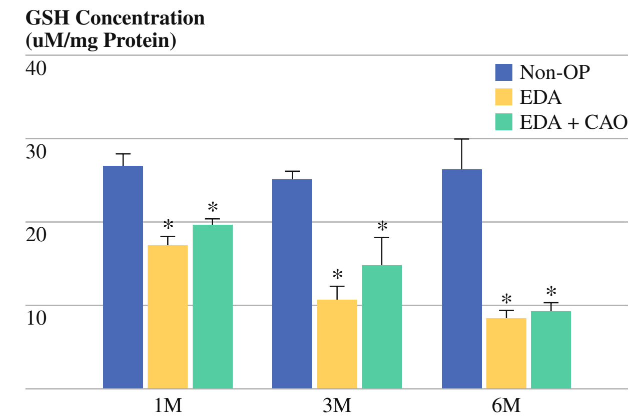

FIG. 5 Effect of *Curcuma aromatica* oil on Glutathione in rat esophageal epithelium after EDA. *Non-OP* nonoperation, *EDA* esophagoduodenal anastomosis, *CAO Curcuma aromatica* oil, *M* month. Data are mean ± SD. * $p < 0.05$ compared with non-OP group; $^{\#}$ $p < 0.05$ compared with EDA group

decreased the incidences of intestinal metaplasia and EAC induced by duodenogastroesophageal reflux. We also observed that *Curcuma aromatica* oil reduced the abnormal hyperproliferation of esophageal epithelial cells corresponding to the reduced incidence of intestinal metaplasia and EAC. Furthermore, *Curcuma aromatica* oil ameliorated the loss of MnSOD, induced by oxidative injury, which could be an important mechanism associated with developments of intestinal metaplasia and EAC.

Curcuma aromatica has long been used as a dietary pigment and spice in Asian countries, such as Indian, Thailand, and China.[23] It also has been used as a traditional Asian medicine to treat gastrointestinal upset, arthritic pain, and cancer since ancient times.[24] The principal biological active ingredients of *Curcuma aromatica* are consistent with two fractions: crystallization and volatile oil. The crystallization fraction is normally obtained by alcohol extraction of the powdered curcuma rhizome, which mainly contains these related cucuminoids: curcumin, bisdesmethoxycurcumin, and desmethoxycurcumin. The volatile fraction, extracted by steam distillation, is an oil that contains a different spectrum of components related to sesquiterpenoids, including curcumenone, curcumenol, neocurdione, curdione, isocurcumenol, furanodienone, curcumol, germacrone, curzerene, furanodiene, and β-elemene.[25–27] At present, numerous in vitro and animal studies have shown that curcuminoids, especially curcumin, exhibit significant anticarcinogenic effects[28]; however, the anticancer properties of the volatile oil of *Curcuma aromatica* have received less attention, only from a few animal studies.[29,30]

In this study, we reproduced a rat EDA model, which has been established in our previous studies, to induce esophagitis, metaplasia, and EAC, and treated the animals with *Curcuma aromatica* oil after EDA. An important issue addressed in this study is that administration of *Curcuma aromatica* oil prevented esophageal epithelium from EDA-induced reflux esophagitis, and decreased the degree of reflux esophagitis, Barrett's metaplasia, and EAC. The esophageal protective effect of *Curcuma aromatica* oil against the EDA reflux esophagitis could be explained by its anti-inflammatory property. These properties are supported by the observations from both clinical and animal studies regarding the anti-inflammatory effect of these sesquiterpenoids in *Curcuma aromatica* oil.[31] The anti-inflammatory activity of *Curcuma aromatica* oil could be accounted for by a number of mechanisms. These include inhibition of cyclooxygenase-2 (COX-2), down-regulation of prostaglandin E_2, suppressing the secretions of both tumor necrosis factor-alpha and interleukin-1β, inhibiting activation of the transcription factors NF-KappaB.[32–35] In addition to the activity of anti-inflammatory, the ingredients in *Curcuma aromatica* oil have antioxidant activities,[36] therefore, the esophageal protective effect by *Curcuma aromatica* oil could response through an antioxidant mechanism.

The involvement of oxidative stress in GERD and its complications, such as BE and EAC, has been pursued actively during the last decade. Evidence indicates that ROS is closely associated with the severity of esophagitis[37,38] and EAC.[39] Several animal studies have indicated that oxidative injury plays a critical role during the transition from esophagitis to BE to EAC.[40–43] In this study, the increased levels of lipid peroxide and 8-OH-dG along with increase of apoptosis after EDA were observed, which suggested that oxidative insult are involved in the initiation of reflux-induced esophageal cell death. The increased proliferation rate within the EDA esophageal epithelium is considered an adaptive response to cell loss in reflux esophagitis to maintain epithelial thickness[44,45]; however, it also is an important mechanism for the initiation of BE and EAC. We explored the role of *Curcuma aromatica* oil against esophageal oxidative damage, and the results indicated that *Curcuma aromatica* oil can not only decrease the levels of both lipid peroxide and 8-OH-dG but also decrease the high rates of both proliferation and apoptosis in the EDA esophageal epithelium. Although the active antioxidant principles in curcuma species have been identified as curcumin,[46] the ingredients in *Curcuma aromatica* oil also have some antioxidant activities, which is supported by these evidences: 1) xanthorrhizol, a natural sesquiterpenoid isolated from the rhizome of curcuma showed antioxidant activity[47]) the aqueous extracts other than curcumin isolated from curcuma species have shown antioxidative activity[48,49]) it has been demonstrated that curcumin is poorly absorbed after ingestion; however, diet supplementation with curcuma species or its extracts to animal still showed antioxidant activity,[50–52] therefore, the antioxidant activity could be from the ingredients in *Curcuma aromatica* oil.

It has been demonstrated that the generation of the superoxide anion is the main free radical involved in mucosal damage in reflux esophagitis,[53–55] and superoxide anion is mainly scavenged by SOD. In this study, we found a decreased MnSOD protein expression and enzymatic activity at 1 month after EDA, and MnSOD but not Cu/ZnSOD contributes to the loss of total SOD. Administration of *Curcuma aromatica* oil can prevent the loss of MnSOD; however, *Curcuma aromatica* oil did not affect the status of endogenous antioxidants, such as Cu/ZnSOD, GSH, and catalase. This is similar to our previous experiments in evaluating the administration of a MnSOD mimic in which the lost of MnSOD was prevented and lead to a reduced incidence of EAC in those animals.[56] However, because the oral administration of this mimic is not possible, our laboratory has begun to investigate potentially naturally occurring compounds that may have similar MnSOD effects. Evidences indicated that superoxide anion might act as endogenous carcinogens by damaging DNA and initiating cells to transform if oxidative mutations are not correctly repaired under the compromised condition of SOD. ROS not only acts as endogenous carcinogens by damaging DNA and initiating cells to transformation,[57] but also participates in mitogenic signal transduction in tumor promotion.[58] Therefore, the protective effects of *Curcuma aromatica* oil against oxidative injury could be directly through its preservation of MnSOD thereby preventing metaplasia and EAC.

Interestingly, a progressive increase of MnSOD is found after EDA 3 and 6 months, which corresponds to our previous observations. This is seen in both groups, whether treated with *Curcuma aromatica* oil or not, and there also are increases in apoptosis and lipid peroxide levels in the later EDA stage (6 month) with *Curcuma aromatica* oil treatment. The possible reasons could be that MnSOD was induced to a higher level to compensate the relative insufficient MnSOD by continuous reflux extending during a much longer period of time.[59,60] In this regard, a relative higher prooxidant action could take place in the EDA esophageal epithelium,[61] and the increased lipid peroxide level and apoptosis may favor the anticarcinogenesis effect of *Curcuma aromatica* oil. However, it should be noted that some other unknown mechanisms also could be involved in these cellular events in the EDA later stage. The role of *Curcuma aromatica* oil in the later stage of EDA and its anti-malignant transformation need to be further addressed in future studies.

CONCLUSIONS

Our results suggest that administration with *Curcuma aromatica* oil is effective to prevent Barrett's metaplasia and esophageal adenocarcinoma in an EDA rat model. The antioxidative effect of *Curcuma aromatica* oil is closely related to preserving MnSOD in response to the reflux injury at the early EDA stage, and the inhibition of oxidative stress by *Curcuma aromatica* oil is critical for the prevention against intestinal metaplasia and cancer.

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